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(54) **MULTI-FUNCTIONAL LIGHT MODULE FOR AN EXTERIOR MIRROR**

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This patent is subject to a terminal disclaimer.

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F21S 43/14 (2018.01)
(52) **U.S. Cl.**
CPC **F21S 43/237** (2018.01); **F21S 43/14** (2018.01)

(58) **Field of Classification Search**
CPC **F21S 54/14**
See application file for complete search history.

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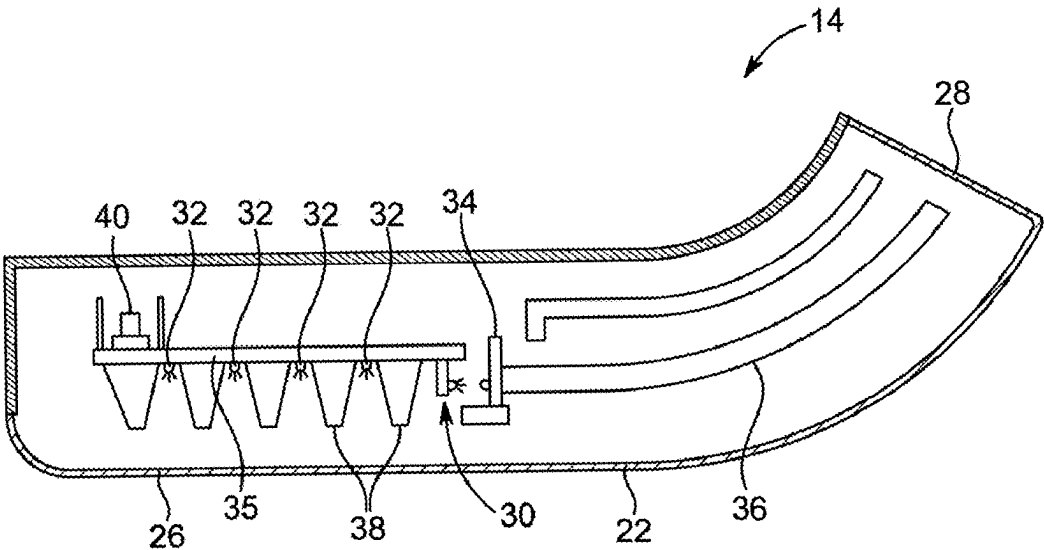
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(57) **ABSTRACT**

The present disclosure refers to a light module for an exterior mirror of a vehicle, comprising a housing, a lens attached to the housing and including at least a first lens section that is configured to direct light in a first direction and a second lens section that is configured to direct light in a second direction, a first light emitting diode (LED) mounted within the housing, a plurality of second LEDs mounted within the housing and configured to emit light through the first lens section, and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section.

18 Claims, 6 Drawing Sheets



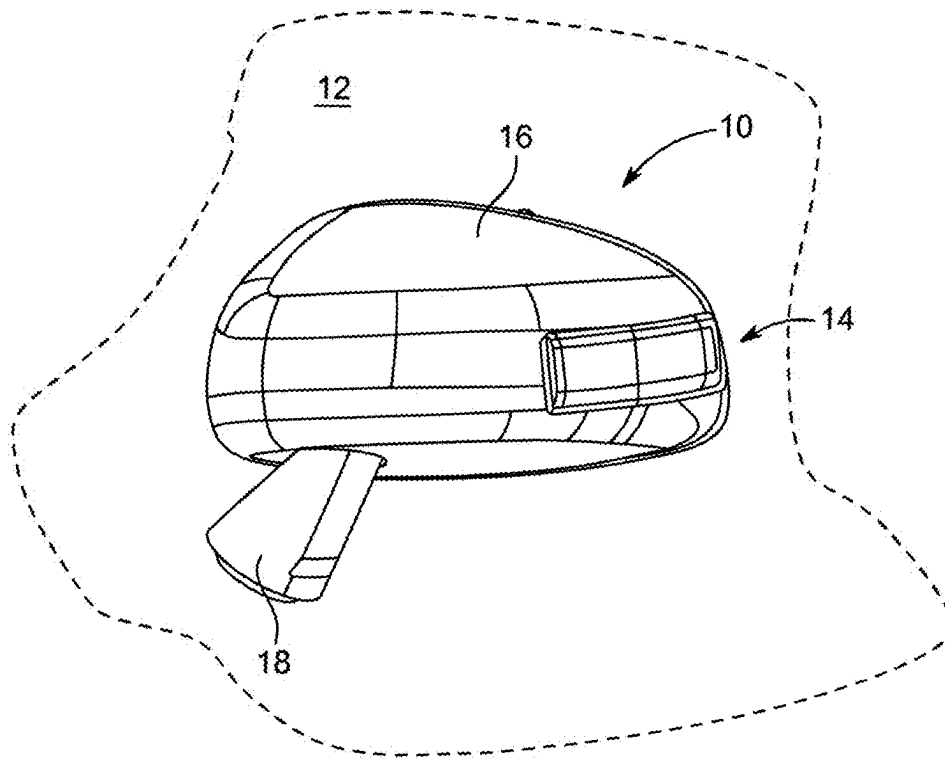
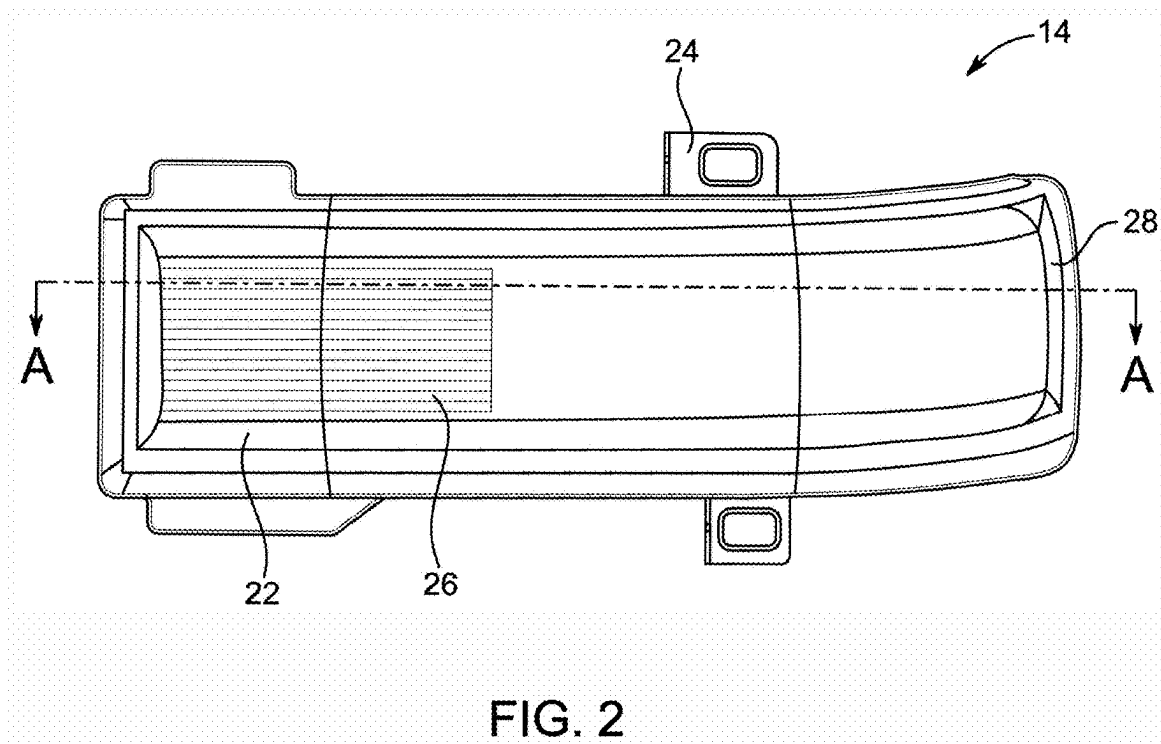


FIG. 1



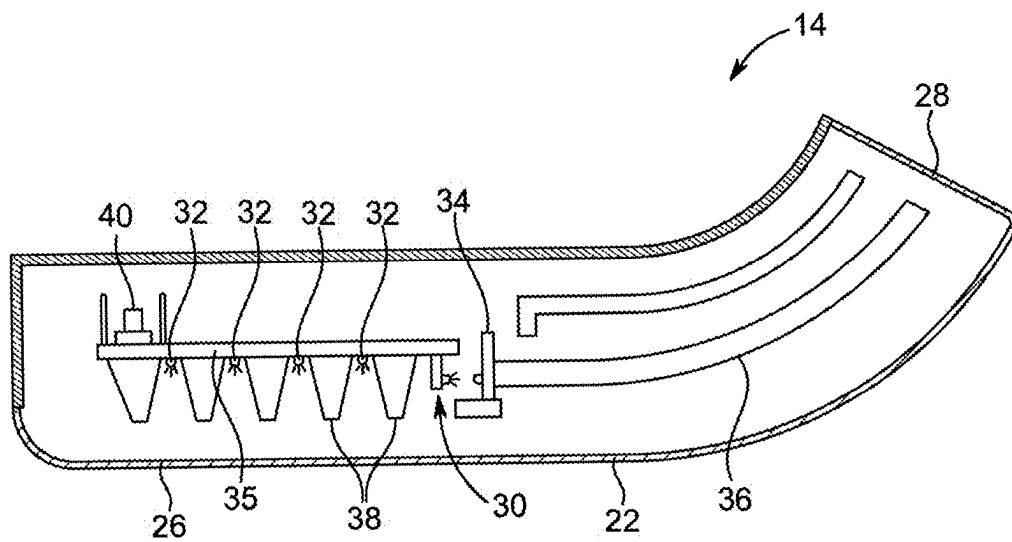
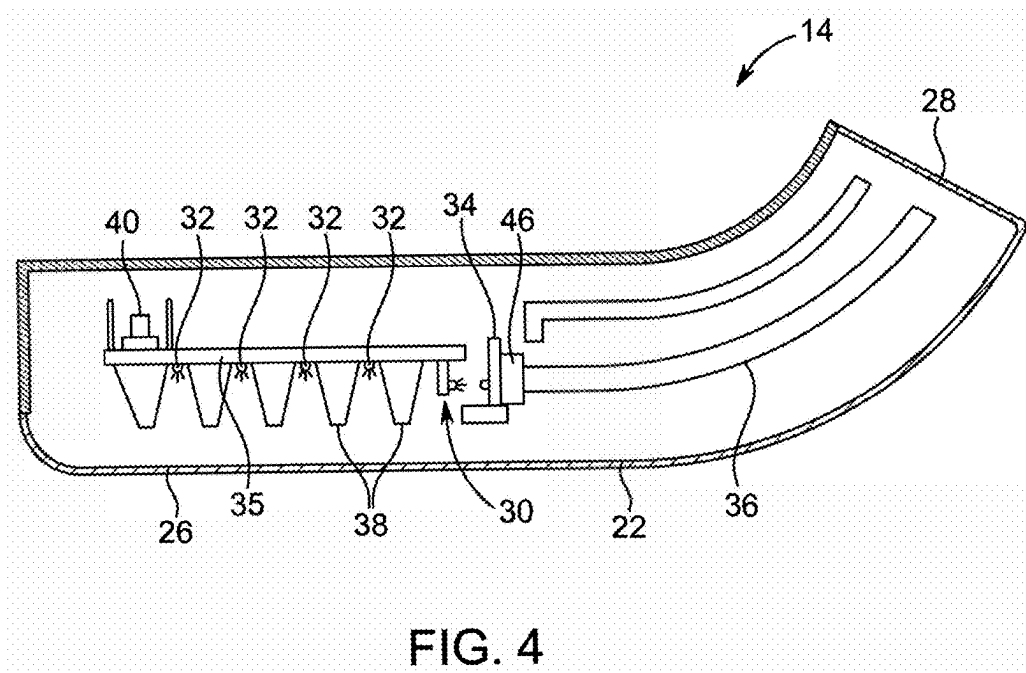


FIG. 3



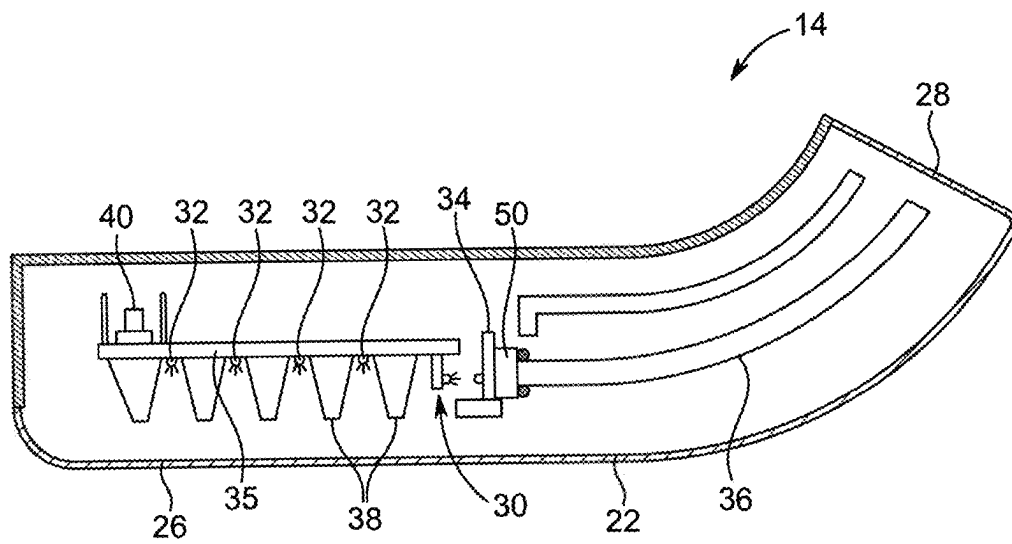


FIG. 5

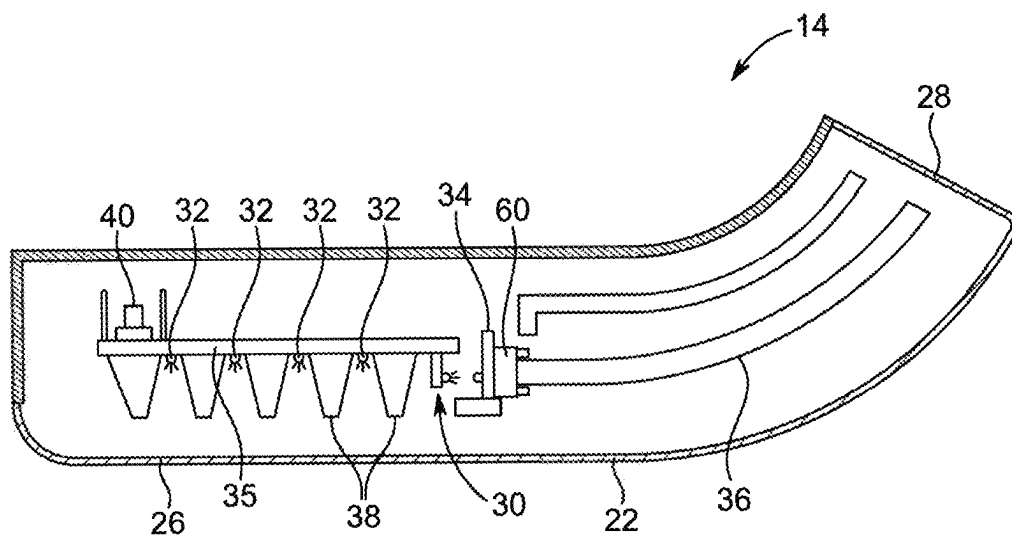


FIG. 6

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MULTI-FUNCTIONAL LIGHT MODULE FOR AN EXTERIOR MIRROR

CROSS-REFERENCE TO RELATED APPLICATION

This is a continuation of U.S. patent application Ser. No. 18/584,143, filed on Feb. 22, 2024, the entirety of which is incorporated herein by reference.

FIELD

The present disclosure relates generally to a multi-functional light module for a vehicle, in particular for a rearview device of a motor vehicle.

BACKGROUND

The statements in this section provide background information relating to the present disclosure and are not intended to constitute prior art.

Vehicles, such as passenger cars, vans, and trucks, typically include rear view mirror assemblies. These assemblies increasingly include light modules to provide various lighting functions for the vehicle, such as security lights, running lights, puddle lights, turn signal lights, and spotlights. These lighting functions packaged in a single light module are often used for driver notification and for security and safety functions. Packaging multiple lighting functions in a mirror assembly can create difficulty in utilizing a mirror design across different countries with different government regulations relating to the type and color of each light function. In some cases it may be desirable to have the light functions differ across countries. Due to government regulations, light emitting from the mirror assembly may be required to maintain a particular color for all light associated with a given operation. When the operations vary from product line to product line, communizing one light source in a multi-functional light module becomes more challenging. It is desirable to develop a system where the light system may be communized and the light emitted may be modified to meet regulations.

SUMMARY

The present disclosure provides a light module for an exterior mirror of a vehicle, comprising a housing, a lens attached to the housing and including at least a first lens section that is configured to direct light in a first direction and a second lens section that is configured to direct light in a second direction, a light module printed circuit board (PCB) mounted within the housing, a first light emitting diode (LED) attached to the light module PCB, a plurality of second LEDs attached to the light module PCB and configured to emit light through the first lens section, and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section.

In embodiments, light emitted through the first lens section is white light and light emitted through the second lens section is non-white light. For example, the first LED and the plurality of second LEDs may be all white light emitting LEDs, where white light emitted from the first LED is filtered to provide the non-white light emitted through the second lens section. In embodiments, the light emitted through the second lens section is amber light. In embodiments, the light emitted through the first lens section is

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directed in a forward-facing direction with respect to a direction of forward movement of the vehicle.

In embodiments, the light module may further include a plurality of light-directing structures configured to direct light from the plurality of second LEDs through the first lens section. In embodiments, the plurality of light-directing structures include reflective surfaces that direct light from the plurality of second LEDs in a direction perpendicular to a surface of the light module PCB and through the first lens section.

In embodiments, the light emitted through the second lens section is directed in a side- and/or rear-facing direction with respect to a direction of forward movement of the vehicle. In embodiments, the light emitted through the second lens section provides a running light and/or turn signal function for the vehicle. In embodiments, the light emitted through the second lens section is in a normally-on state to provide the running light function and turns off and on to provide the turn signal function for the vehicle.

In embodiments, the light module may further include an optical coupler configured to receive light from the first LED and direct light into the light pipe. In embodiments, the optical coupler includes a filter that filters the spectrum of the white light emitted from the first LED to provide the non-white light emitted through the second lens section. Embodiments may further include a filter attached to the optical coupler that filters the spectrum of the white light emitted from the first LED to provide the non-white light emitted through the second lens section.

In other embodiments, the light pipe filters the white light emitted from the first LED to provide the non-white light emitted through the second lens section.

An embodiment of the present disclosure provides a light module for an exterior mirror of a vehicle, comprising a housing, a lens attached to the housing and including at least a first lens section that is configured to direct light in a forward-facing direction with respect to a direction of forward movement of the vehicle and a second lens section that is configured to direct light in a one or more directions different than the forward-facing direction, a light module printed circuit board (PCB) mounted within the housing, a first light emitting diode (LED) attached to the light module PCB, a plurality of second LEDs attached to the light module PCB and configured to emit light through the first lens section, and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section. In embodiments, the light emitted through the second lens section is directed in a side- and/or rear-facing direction with respect to the direction of forward movement of the vehicle. In embodiments, the light emitted through the second lens section provides a running light and/or turn signal function for the vehicle.

Another embodiment of the present disclosure provides a light module for an exterior mirror of a vehicle, comprising a housing, a lens attached to the housing and including at least a first lens section that is configured to direct white light in a forward-facing direction with respect to a direction of forward movement of the vehicle and a second lens section that is configured to direct non-white light in a one or more directions different than the forward-facing direction, a light module printed circuit board (PCB) mounted within the housing, a first light emitting diode (LED) attached to the light module PCB, a plurality of second LEDs attached to the light module PCB and configured to emit light through the first lens section, and a light pipe mounted within the housing and configured to receive light

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emitted from the first LED and emit light through the second lens section. In embodiments, the first LED is a white light emitting LED, and wherein white light emitted from the first LED is filtered to provide the non-white light emitted through the second lens section.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing summary, as well as the following detailed description, will be better understood when read in conjunction with the appended drawings. For the purpose of illustration, certain examples of the present disclosure are shown in the drawings. It should be understood, however, that the present disclosure is not limited to the precise arrangements and instrumentalities shown. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate an implementation of system, apparatuses, and methods consistent with the present disclosure and, together with the detailed description, serve to explain advantages and principles consistent with the present disclosure, wherein:

FIG. 1 illustrates an example perspective view of an exterior rearview mirror assembly including a multi-function light module;

FIG. 2 illustrates a front perspective view of a multi-function light module;

FIG. 3 illustrates a sectional view of the multi-function light module; and

FIGS. 4-6 illustrate examples of the multi-functional light module with an optical filter to provide non-white light.

DETAILED DESCRIPTION

The following description is exemplary in nature and is not intended to limit the present disclosure, application, or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

FIG. 1 illustrates an example perspective view of a rearview mirror assembly 10 including a light module 14. The exterior rearview mirror assembly 10 further includes a mirror head 16 and a mirror base 18 and is mounted on a vehicle surface 12. The mirror head 16 is configured to carry at least a reflective element (not shown). In addition, the mirror head 16 may carry at least one camera. In an embodiment, the rearview device may be in the form of a camera pod with a camera head carrying at least one camera.

The rearview assembly 10 in FIG. 1 is mounted to an A-pillar, but may be mounted in other locations on a motor vehicle in other embodiments. The rearview mirror assembly 10 may also have other functions, such as power-folding capability, breakaway capability, telescoping capability, and/or other light assemblies mounted on the mirror head 16 or the mirror base 18.

FIG. 2 is a perspective view of the light module 14. The light module 14 has a lens 22 which may contain optics to direct or modify a light source mounted behind the lens 22. A housing 24 is used to mount the components of the light module 14 and provide a mounting arrangement for fastening the light module 14 to the mirror head 16. The lens 22 is connected or joined to the housing 24 (e.g., by welding, adhesive, or other known fastening method) and has an inner surface and an outer surface which provide a viewing surface that is illuminated by the light module 14 in use. The lens 22 may include at least a first lens section 26 that is configured to direct light in a first direction and a second lens section 28 that is configured to direct light in a second

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direction. The first lens section 26 may, for example, be configured to direct light in a forward-facing direction with respect to a direction of forward movement of the vehicle, for example to provide a forward-facing spotlight for the vehicle. The second lens section 28 may, for example, be configured to direct light in a side- and/or rear-facing direction with respect to a direction of forward movement of the vehicle, for example to provide a running light and/or turn signal function for the vehicle.

FIG. 3 illustrates a sectional view of the light module 14 utilizing the section A-A illustrated in FIG. 2. As shown in FIG. 3, the light module 14 includes a first light emitting diode (LED) 30 and a plurality of second LEDs 32 all mounted to and controlled through the same printed circuit board (PCB) 35. Also attached to the light module PCB 35 are a plurality of light-directing structures 38 and an connector device 40 for electrically connecting the light module PCB 35 to a wire harness for the rearview mirror assembly 10. The light-directing structures 38 may, for example, include reflective surfaces that direct light from the plurality of second LEDs 32 in a direction perpendicular to the surface of the light module PCB 35 and through the forward-facing portion 26 of the light module lens 22. In this way, the second LEDs 32 may provide a forward-facing light function for the light module 14, such as a forward-facing spotlight. For example, in embodiments, the second LEDs 32 may include 2-4 white LEDs having a light intensity that is selected to provide a spotlight function.

As illustrated, the first LED 30 on the light module PCB 35 may be directed in a different direction than the plurality of second LEDs 32, such as in a direction parallel to the surface of the PCB 35. In the illustrated embodiment, the light module 14 further includes an optical coupler 34 located adjacent the first LED 30 and configured to direct light from the first LED 30 into a light pipe 36, such as an optic fiber or other mechanism for transporting and emitting light. The light pipe 36 is configured to receive light from the first LED 30 and emit the light in a direction different than the light emitted from the plurality of second LEDs 32, such as through the side- and/or rear-facing portion 28 of the light module lens 22. In this way, the first LED 30 and associated light pipe 36 may be used, for example, to provide a side- and/or rear-facing function for the light module 14, such as a turn signal light.

In order to provide a specified color for the side-facing light function, the optical coupler 34 may include or have an attached colored lens to modify the color of the first LED 30. For instance, in an embodiment, the first LED 34 and the plurality of second LEDs 32 may all be white light LEDs, and a color filter 34 may be provided to modify the color of the light emitted by the first LED 30. For example, if the first LED 30 is utilized to provide a turn signal function, then an amber-colored lens 34 may be included to change white light emitted by the first LED 30 to amber light emitted from the light pipe 36 through a side-facing portion of the light module lens 22. In an embodiment, the light emitted by the first LED 30 may be filtered to provide the non-white light (e.g., amber light) emitted through the second lens section 28. In another embodiment, the light pipe 36 may be configured (e.g., colored) to filter white light emitted from the first LED 30 to provide the desired color of non-white light emitted through the second lens section 28. In other embodiments, for example as shown in FIGS. 4-6, an optical filter may be included between the first LED 30 and light pipe 36 to provide the desired light color.

In one embodiment, the light emitted from the first LED 30 and light pipe 36 may be controlled (e.g., by a controller

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coupled to the connector device 40) to provide non-white light through the second lens section 28 that is in a normally-on state to provide the running light function for the vehicle, and that turns off and on to provide the turn signal function for the vehicle.

As shown in FIG. 4, the light module 14 may further include an optical filter 46 (e.g., a colored lens) to modify the color of the first LED 34. For instance, in an embodiment, the first LED 34 and plurality of second LEDs 30 may all be white light LEDs, and a color filter 46 may be provided to modify the color of the light emitted by the first LED 34. For example, if the first LED 34 is utilized to provide a turn signal function, then an amber-colored lens 46 may be included to change white light emitted by the first LED 34 to amber light emitted from the light pipe 36 through the side-facing section 28 of the light module lens 22.

In embodiments, an optical filter (e.g., colored lens) may be welded or clipped to the light pipe 35. For example, FIG. 5 illustrates an embodiment with an optical filter 50 that is welded to the light pipe 36, and FIG. 6 another embodiment with an optical filter 60 that is clipped to the light pipe 36.

It will be appreciated by those skilled in the art that changes could be made to the embodiments described above without departing from the broad inventive concept thereof. It is understood, therefore, that the invention disclosed herein is not limited to the particular embodiments disclosed, and is intended to cover modifications within the spirit and scope of the present invention.

REFERENCE SIGNS

10 rearview mirror assembly
12 vehicle surface
14 light module
16 mirror head
18 mirror base
22 lens
24 housing
26 first lens section
28 second lens section
30 first LED
32 plurality of second LEDs
34 optical coupler
35 light module PCB
36 light pipe
38 light-directing structures
40 electrical connector
46 optical filter
50 welded optical filter
60 clipped optical filter

It is claimed:

1. A light module for an exterior mirror of a vehicle, comprising:

a housing;
a lens attached to the housing and including at least a first lens section that is configured to direct light in a first direction and a second lens section that is configured to direct light in a second direction;
a first light emitting diode (LED) mounted within the housing;
a plurality of second LEDs mounted within the housing and configured to emit light through the first lens section; and
a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section,

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wherein the first LED and the plurality of second LEDs are all white light emitting LEDs, and wherein white light emitted from the first LED is filtered to provide amber light that is emitted through the second lens section.

2. The light module of claim 1, further comprising a light module printed circuit board (PCB) mounted within the housing, wherein the first LED is attached to the light module PCB.

3. The light module of claim 1, further comprising a light module printed circuit board (PCB) mounted within the housing, wherein the plurality of second LEDs are attached to the light module PCB.

4. The light module of claim 3, wherein the light emitted through the first lens section is directed in a forward-facing direction with respect to a direction of forward movement of the vehicle.

5. The light module of claim 4, further comprising a plurality of light-directing structures configured to direct light from the plurality of second LEDs through the first lens section.

6. The light module of claim 5, wherein the plurality of light-directing structures include reflective surfaces that direct light from the plurality of second LEDs in a direction perpendicular to a surface of the light module PCB and through the first lens section.

7. The light module of claim 1, wherein the light emitted through the second lens section is directed in a side- and/or rear-facing direction with respect to a direction of forward movement of the vehicle.

8. The light module of claim 7, wherein the light emitted through the second lens section provides a running light and/or turn signal function for the vehicle.

9. The light module of claim 8, wherein the light emitted through the second lens section is in a normally-on state to provide the running light function and turns off and on to provide the turn signal function for the vehicle.

10. The light module of claim 1, further comprising:
an optical coupler configured to receive light from the first LED and direct light into the light pipe.

11. The light module of claim 10, wherein the optical coupler includes a filter that filters the spectrum of the white light emitted from the first LED to provide the non-white light emitted through the second lens section.

12. The light module of claim 10, further comprising a filter attached to the optical coupler that filters the spectrum of the white light emitted from the first LED to provide the non-white light emitted through the second lens section.

13. The light module of claim 1, wherein the light pipe filters the white light emitted from the first LED to provide the non-white light emitted through the second lens section.

14. A light module for an exterior mirror of a vehicle, comprising:

a housing;
a lens attached to the housing and including at least a first lens section that is configured to direct light in a forward-facing direction with respect to a direction of forward movement of the vehicle and a second lens section that is configured to direct light in a one or more directions different than the forward-facing direction;
a first light emitting diode (LED) mounted within the housing;
a plurality of second LEDs mounted within the housing and configured to emit light through the first lens section; and

a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section,

wherein the first LED and the plurality of second LEDs are all white light emitting LEDs, and wherein white light emitted from the first LED is filtered to provide amber light that is emitted through the second lens section. 5

15. The light module of claim **14**, further comprising a light module printed circuit board (PCB) mounted within the housing, wherein the first LED is attached to the light module PCB. 10

16. The light module of claim **14**, further comprising a light module printed circuit board (PCB) mounted within the housing, wherein the plurality of second LEDs are attached to the light module PCB. 15

17. The light module of claim **14**, wherein the light emitted through the second lens section is directed in a side-and/or rear-facing direction with respect to the direction of forward movement of the vehicle. 20

18. The light module of claim **17**, wherein the light emitted through the second lens section provides a running light and/or turn signal function for the vehicle.

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